

Given: $\vec{F}_B = \begin{bmatrix} F_{xB} \\ F_{yB} \\ F_{zB} \end{bmatrix}$, $\vec{M}_B = \begin{bmatrix} M_{xB} \\ M_{yB} \\ M_{zB} \end{bmatrix}$, $\vec{Q}_i = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$

$\phi_i = \theta_i = \psi_i = 0$

Nav. Frame: $\overset{x}{\text{UP}}, \overset{y}{\text{East}}, \overset{z}{\text{North}}$

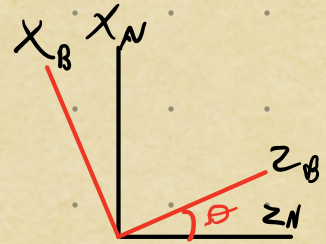
Body Frame: $\overset{x}{\text{Roll}}, \overset{y}{\text{aligned w/ East}}, \overset{z}{\text{aligned w/ North}}$

Rot. order: ZYX, passive rot.

δ : Gimbal in pitch axis; γ : Gimbal in yaw axis

Find: $\ddot{X}_N, \ddot{Y}_N, \ddot{Z}_N, \ddot{\phi}_{NB}, \ddot{\theta}_{NB}, \ddot{\psi}_{NB}$

$\dot{X}_N, \dot{Y}_N, \dot{Z}_N, \dot{\phi}_{NB}, \dot{\theta}_{NB}, \dot{\psi}_{NB}$



Assume: m is constant; $I_1 = I_2 = I$; model sys. as particle

$$\vec{F}_N = \vec{T}_N - m \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}_N; \quad \vec{T}_N = R_{NB} \vec{T}_B = R_{BN}^T \vec{T}_B$$

$$\vec{T}_B = ||T|| \begin{pmatrix} \cos \delta \cos \gamma \\ \sin \gamma \\ -\sin \delta \end{pmatrix}$$

(BN is passive rot. in neg. dir $\rightarrow$ active rot)

$$\vec{T}_N = \begin{pmatrix} \cos \psi \cos \theta \\ \cos \psi \sin \theta \sin \phi - \cos \phi \sin \psi \\ \sin \theta \sin \psi + \cos \phi \cos \psi \sin \theta \end{pmatrix} \begin{pmatrix} \cos \delta \sin \gamma \\ \cos \phi \cos \gamma + \sin \phi \sin \gamma \sin \theta \\ \cos \phi \sin \gamma \sin \theta - \cos \psi \sin \phi \end{pmatrix}$$

$$\begin{pmatrix} \sin \theta \\ \cos \theta \sin \phi \\ \cos \theta \cos \phi \end{pmatrix}_{NB} \begin{pmatrix} \cos \delta \cos \gamma \\ \sin \gamma \\ -\sin \delta \end{pmatrix}_B$$

$$\ddot{\vec{r}}_N = \frac{\vec{T}_N}{m} - \begin{pmatrix} g \\ 0 \\ 0 \end{pmatrix} = \frac{\cos \delta \cos \gamma \|\vec{T}\|}{m} \begin{pmatrix} \cos \psi \cos \theta \\ (\cos \psi \sin \phi \sin \theta - \cos \phi \sin \psi) \\ (\sin \phi \sin \psi + \cos \phi \cos \psi \sin \theta) \end{pmatrix} +$$

$$\frac{\sin \gamma \|\vec{T}\|}{m} \begin{pmatrix} \cos \psi \sin \phi \sin \theta - \cos \phi \sin \psi \\ \cos \phi \cos \psi + \sin \phi \sin \psi \sin \theta \\ \cos \phi \sin \psi \sin \theta - \cos \psi \sin \theta \end{pmatrix} +$$

$$-\frac{\sin \delta \|\vec{T}\|}{m} \begin{pmatrix} \sin \phi \sin \psi + \cos \phi \cos \psi \sin \theta \\ \cos \phi \sin \psi \sin \theta - \cos \psi \sin \theta \\ \cos \phi \cos \theta \end{pmatrix} - \begin{pmatrix} g \\ 0 \\ 0 \end{pmatrix}$$

$$\ddot{\chi}_N = \frac{\|\vec{T}\|}{m} (\cos \delta \cos \gamma \cos \psi \cos \theta + \sin \gamma \cos \psi \sin \phi \sin \theta - \sin \gamma \cos \phi \sin \psi +$$

$$- \sin \gamma \sin \phi \sin \psi - \sin \gamma \cos \phi \cos \psi \sin \theta) - g \quad (1)$$

$\dot{V}_{\chi N}$

$$\ddot{Y}_N = \frac{\|\vec{T}\|}{m} (\cos \delta \cos \gamma \cos \psi \sin \phi \sin \theta - \cos \delta \cos \gamma \cos \phi \sin \psi +$$

$$\sin \gamma \cos \phi \cos \psi + \sin \gamma \sin \phi \sin \psi \sin \theta +$$

$$- \sin \delta \cos \phi \sin \psi \sin \theta + \sin \delta \cos \psi \sin \theta) \quad (2)$$

$\dot{V}_{Y N}$

$$\ddot{Z}_N = \frac{\|\vec{T}\|}{m} (\cos \delta \cos \gamma \sin \phi \sin \psi + \cos \delta \cos \gamma \cos \phi \cos \psi \sin \theta +$$

$$\sin \gamma \cos \phi \sin \psi \sin \theta - \sin \gamma \cos \psi \sin \phi +$$

$$- \sin \delta \cos \phi \cos \theta) \quad (3)$$

$\dot{V}_{Z N}$

$$V_{\chi N} = \dot{\chi}_N \quad (4)$$

$$V_{Y N} = \dot{Y}_N \quad (5)$$

$$V_{Z N} = \dot{Z}_N \quad (6)$$

$$\sum \vec{M}_B = \vec{M}_B = \vec{r}_{TB \rightarrow CB} \times \vec{T}_B = \begin{pmatrix} L \\ 0 \\ 0 \end{pmatrix}_B \times ||T|| \begin{pmatrix} \cos\delta \cos\gamma \\ \sin\gamma \\ -\sin\delta \end{pmatrix}$$

$$\vec{M}_B = ||T|| \begin{pmatrix} 0 \\ L \sin\gamma \\ L \sin\delta \end{pmatrix}$$

$$\vec{M}_{xB} = 0; \quad \vec{M}_{yB} = ||T|| L \sin\gamma; \quad \vec{M}_{zB} = ||T|| L \sin\delta$$

$$\dot{\omega}_{xB} = 0; \quad \dot{\omega}_{yB} = \begin{pmatrix} 0 \\ \frac{||T|| L \sin\gamma}{I} \\ 0 \end{pmatrix}; \quad \dot{\omega}_{zB} = \begin{pmatrix} 0 \\ 0 \\ \frac{||T|| L \sin\delta}{I} \end{pmatrix}$$

$$\dot{\omega}_N = R_{NB} \dot{\omega}_B = R_{NB} (\dot{\omega}_{xB} + \dot{\omega}_{yB} + \dot{\omega}_{zB})$$

$$= \dot{\omega}_N +$$

$$\frac{||T|| L \sin\gamma}{I} \begin{pmatrix} \cos\theta \sin\psi \\ \cos\phi \cos\psi + \sin\phi \sin\psi \cos\theta \\ \cos\phi \sin\psi \sin\theta - \cos\psi \sin\phi \end{pmatrix}_N +$$

$$\frac{||T|| L \sin\delta}{I} \begin{pmatrix} -\sin\theta \\ \cos\theta \sin\phi \\ \cos\phi \cos\theta \end{pmatrix}$$

$$\dot{\omega}_{xN} = \frac{||T|| L}{I} (\sin\gamma \cos\theta \sin\psi - \sin\delta \sin\theta) \quad (7)$$

$$\dot{\omega}_{yN} = \frac{||T|| L}{I} (\sin\gamma \cos\phi \cos\psi + \sin\gamma \sin\phi \sin\psi \cos\theta + \sin\delta \cos\theta \sin\phi) \quad (8)$$

$$\dot{\omega}_{ZN} = \frac{1/|\mathbf{T}|}{I} \left(\sin\gamma \cos\phi \sin\psi \sin\theta - \sin\gamma \cos\psi \sin\phi + \sin\delta (\cos\phi \cos\theta) \right) \quad (9)$$